

Form and Meaning in BERT Embeddings

Analyzing the verbal 'te'-infinitival complement clause with optional complementizer 'om' in Dutch and multilingual BERT models

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Introduction

The goal is to study how well linguistic constructions are encoded by Large Language Models.

The specific Dutch construction is the verbal 'te'-infinitival complement clause with the optional complementizer 'om'. Some sentences containing this construction are:

- *Zij beloven (om) te helpen.* (They promise to help.)
- *Ik ben vergeten (om) eieren te kopen.* (I forgot to buy eggs.)

These sentences can be written with and without the word 'om', and the inclusion or exclusion of this word does not change the meaning of the sentence.

BERT models

RobBERT-2023	monolingual	based on RoBERTa
BERTje	monolingual	based on BERT
EuroBERT	multilingual	based on BERT
mBERT	multilingual	based on BERT

Methodology

- The data consists of sentences containing three different constructions
 - 'Om te' sentences, containing the verbal 'te'-infinitival complement clause with the complementizer 'om'
 - *Jij besluit om te reizen.*
 - 'Te' sentences, containing the verbal 'te'-infinitival complement clause without the complementizer 'om'
 - *Jij besluit te reizen.*
 - 'Niet te' sentences, containing the verbal 'te'-infinitival complement clause preceded by the word 'niet'
 - *Jij besluit niet te reizen.*
- These sentences are divided in two groups:

Group that differs only in form
'om te' sentences + 'te' sentences

Group that differs in form and meaning
'te' sentences + 'niet te' sentences

- These sentences are encoded into word embeddings by the different BERT models, these word embeddings have 1024 or 768 dimensions
- Sentence embeddings are calculated from these word embeddings by mean pooling, which is calculating the average of the word embeddings per sentence
- Cluster coherence of these pre-existing clusters are measured by the following internal cluster analysis methods:

Silhouette Score

Compares the distances between vectors within a cluster to the distances of vectors between clusters.

Davies-Bouldin Index

Compares the distances within a cluster with the distance between the cluster centroids.

Sentences closest to average

The average sentence embedding of the 'om te' sentences is calculated and the top 10 and top 100 sentences closest to this average are found from both the 'om te' sentences and 'te' sentences. Then the amount of sentences from each group is calculated.

Results

- Better cluster coherence was found in the group that differed in meaning as well as form than in the group that differed only in form
- Monolingual models performed better than multilingual models
- RobBERT-2023 performed best overall, showing a potential advantage for RoBERTa-based architecture
- Generally better cluster coherence found in longer sentences compared to shorter sentences, but BERTje specifically showed better results for shorter sentences than for longer sentences
- The top 10 closest sentences shows a higher than random chance that the closest sentences are from within the 'om te' category for 3 of the models. Expanding to the top 100 shows this for all models.

	Percentage 'om te' sentences in top 10	Percentage 'om te' sentences in top 100
RobBERT-2023	90%	72%
BERTje	80%	66%
EuroBERT	90%	67%
mBERT	30%	61%